

Automated Glacial Lake Mapping and Hazard Assessment in the Hindu Kush–Himalaya Using Deep Learning and PlanetScope Imagery

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Abstract—Glacial lakes in the Hindu Kush–Himalaya (HKH) are rapidly evolving under a warming climate, increasing the risk of glacial lake outburst floods (GLOFs) and downstream impacts. Accurate hazard assessment requires up-to-date and spatially consistent glacial lake inventories, which remain limited by manual mapping efforts. In this study, we present an automated deep learning–based framework for regional-scale glacial lake mapping and hazard assessment using high-resolution PlanetScope imagery. Multiple convolutional and transformer-based segmentation models are evaluated under different spectral input configurations. Results show that RGB-based convolutional models consistently outperform more complex spectral and transformer-based configurations. Using the best-performing model, we generate a 2025 glacial lake inventory comprising 15,353 lakes with a total area of 148.613 km², representing an 8.95% increase relative to 2020. Size- and elevation-dependent analyses reveal widespread formation of small lakes alongside heterogeneous growth among larger systems. A multi-factor hazard assessment identifies 4,519 potentially dangerous glacial lakes, with population exposure estimates indicating that hundreds of millions of people may reside within 20 km downstream. These results demonstrate the potential of deep learning for scalable, risk-informed glacial lake monitoring in high mountain regions.

Index Terms—Glacial lakes, deep learning, semantic segmentation, PlanetScope imagery, hazard assessment, potentially dangerous glacial lakes (PDGLs), Hindu Kush–Himalaya.

I. INTRODUCTION

Glacial lakes in the Hindu Kush–Himalaya (HKH) are rapidly evolving under climate-driven glacier retreat, increasing the risk of glacial lake outburst floods (GLOFs) and associated downstream impacts. Reliable hazard assessment depends on up-to-date and spatially consistent glacial lake inventories; however, existing datasets are largely derived from manual or semi-automated mapping of medium-resolution imagery, limiting their temporal consistency and ability to capture small or short-lived lakes. Recent advances in high-resolution satellite imagery and deep learning-based semantic segmentation provide new opportunities for automated glacial lake mapping, yet the influence of model architecture and spectral configuration remains insufficiently explored, and hazard assessment is often treated separately from lake detection. Moreover, identifying potentially dangerous glacial lakes

(PDGLs) requires integrating geomorphological and hydrological indicators beyond lake extent alone. In this study, we develop an end-to-end deep learning framework for automated glacial lake mapping and hazard assessment across the HKH using PlanetScope imagery. We systematically evaluate multiple convolutional and transformer-based segmentation models under different spectral inputs, generate an updated 2025 glacial lake inventory, quantify changes relative to 2020, and identify PDGLs to estimate downstream population exposure. This work demonstrates a scalable, risk-informed approach for monitoring glacial lakes in high mountain environments.

II. METHODOLOGY

A. Overview

The proposed methodology consists of three main components: (i) preparation of a training dataset from high-resolution PlanetScope imagery using an existing glacial lake inventory, (ii) automated glacial lake mapping using deep learning–based semantic segmentation models, and (iii) hazard-oriented post-processing to identify potentially dangerous glacial lakes (PDGLs) and assess downstream population exposure. The overall design emphasizes scalability, reproducibility, and suitability for regional-scale monitoring in data-scarce high mountain environments.

B. PlanetScope Imagery and Dataset Preparation

We utilized PlanetScope imagery due to its high spatial resolution (approximately 3 m), which is well suited for capturing small and irregularly shaped glacial lakes that are often missed by medium-resolution sensors. For model training, we used the HKH-PK-2020 glacial lake inventory [1] as a reference dataset. PlanetScope scenes corresponding to the inventory were retrieved as both 4-band visual products and 8-band surface reflectance products, with scenes selected to have cloud cover below 20%.

To construct a training dataset suitable for deep learning, each PlanetScope scene was tiled into geo-referenced patches of size 512 × 512 pixels with a 20% spatial overlap. This overlap was introduced to preserve contextual information near tile boundaries and reduce edge artifacts during inference.

Tiles containing less than 60% valid pixels were excluded to avoid artifacts introduced by irregular scene footprints. Glacial lake polygons intersecting each tile were rasterized into binary masks, where lake pixels were labeled as 1 and background pixels as 0. Tiles without any lake pixels were discarded to mitigate extreme class imbalance. Following this filtering process, we obtained 750 image–mask pairs for the 4-band visual product and 850 pairs for the 8-band surface reflectance product.

C. Spectral Input Configurations

Although PlanetScope imagery provides multi-spectral information, the optimal spectral configuration for deep learning–based glacial lake segmentation is not well established. To investigate the influence of spectral inputs on segmentation performance, we evaluated three configurations with increasing spectral complexity. First, RGB-only inputs were considered, as they allow direct use of ImageNet-pretrained encoder backbones and represent a common baseline in semantic segmentation tasks. Second, the full 8-band surface reflectance product was used to assess whether additional spectral information improves discrimination between water bodies and surrounding terrain. Third, a hybrid configuration combining RGB, near-infrared (NIR), and the normalized difference water index (NDWI) was evaluated to explicitly encode water-sensitive spectral characteristics. This experimental design enables a systematic assessment of the trade-offs between spectral richness and model performance.

D. Deep Learning Models and Training Strategy

We evaluated a diverse set of convolutional and transformer-based semantic segmentation models to identify architectures suitable for operational glacial lake mapping. The convolutional models included U-Net [2], U-Net++ [3], U-Net++ with attention decoder [4], DeepLabv3+ [5], and MANet [6], while transformer-based models included TransUNet [7], Swin U-Net [8], and SegFormer [9]. All models were implemented within a unified training framework to ensure fair comparison.

Encoder backbones were selected from commonly used architectures, including ResNet variants, EfficientNet-B0, and MobileNetV2, depending on model compatibility. All models were trained using a binary cross-entropy and Dice loss combination to balance pixel-wise accuracy with shape preservation. The Adam optimizer was employed with learning rates selected based on model stability, and training was conducted for an initial screening phase followed by extended training for top-performing models. Model performance was evaluated using intersection over union (IoU), precision, recall, and F1-score on a held-out validation set.

E. Automated Inference and Inventory Generation

Using the best-performing segmentation model, we developed an automated inference pipeline to generate a regional glacial lake inventory for the year 2025. During inference, PlanetScope scenes were tiled using the same configuration as in training, and predicted lake masks were mosaicked to

produce spatially continuous outputs. Binary masks were then converted into vector polygons through connected-component analysis and geometric simplification. The resulting polygons constitute a GIS-ready glacial lake inventory suitable for spatial analysis and comparison with previous datasets.

F. Computation of Glacial Lake Statistics

To quantify changes in glacial lake characteristics, we computed both aggregate and lake-level statistics for the 2020 and 2025 inventories. Total lake area was obtained by summing the areas of all lake polygons, while lake count was defined as the total number of polygons in each inventory. Lakes were further grouped into predefined size classes to examine size-dependent trends.

Elevation attributes were derived using the Shuttle Radar Topography Mission (SRTM) 1 arc-second digital elevation model. For each lake polygon, elevation was approximated by sampling the DEM at the polygon centroid. Lakes were then grouped into elevation bands of 500 m width, and lake count and total area were computed for each band.

To assess inter-annual changes, absolute differences in lake number and total area between 2020 and 2025 were computed. Relative changes were quantified using the percentage area change (PAC) and annual percentage area change (APAC):

$$PAC = \frac{A_{2025} - A_{2020}}{A_{2020}} \times 100, \quad (1)$$

$$APAC = \frac{PAC}{\Delta t}, \quad (2)$$

where $\Delta t = 5$ years.

G. Lake Matching and Change Classification

To analyze lake-level evolution, polygons from the 2020 and 2025 inventories were spatially matched using an overlap-based approach. Candidate matches were identified using a spatial index, and intersections were evaluated using an intersection-over-union (IoU) metric. Lakes were considered matched if their IoU exceeded a threshold of 0.01.

For matched lakes, relative area change was computed as

$$\Delta A_{\text{rel}} = \frac{A_{2025} - A_{2020}}{A_{2020}}. \quad (3)$$

Lakes were classified as expanded, shrunk, or stable based on predefined relative change thresholds, while lakes present only in the 2025 inventory were classified as new.

H. Identification of Potentially Dangerous Glacial Lakes

Potentially dangerous glacial lakes were identified using a multi-factor hazard assessment framework adapted from established literature [10]. The framework integrates geomorphological and hydrological indicators/factors that are known to influence GLOF susceptibility. These include the mean slope of the parent glacier, the extent of steep non-glacierized terrain surrounding the lake, an approximation of moraine-dam slope near the lake outlet, upstream contributing catchment area, and lake perimeter. A composite hazard index was then computed as a weighted sum of the normalized factors, with

weights adopted from prior studies to reflect their relative importance. Lakes were classified into ordinal hazard classes using Jenks natural breaks, and lakes falling into the high and very high classes were flagged as potentially dangerous.

I. Estimation of Downstream Population Exposure

To assess potential human impacts associated with PDGLs, we estimated downstream population exposure using gridded population data. Population counts were summed within cumulative distance bands downstream of each PDGL. Exposure estimates were aggregated across all PDGLs to provide regional-scale indicators of population at risk.

III. RESULTS

A. Performance of Deep Learning Models

Table I summarizes the best-performing configurations for the RGB input setting.

TABLE I
BEST RESULTS ON RGB DATASET

Model	backbone/hyp param	split	Loss	IoU	F1
U-Net	effnet-b0	train	0.3113	0.6326	0.7384
U-Net	effnet-b0	val	0.3087	0.6379	0.7423
DeepLabv3+	timm-effnet-b0	train	0.3432	0.6034	0.7159
DeepLabv3+	timm-effnet-b0	val	0.3231	0.6250	0.7348
U-Net++	timm-effnet-b0	train	0.2863	0.6543	0.7583
U-Net++	timm-effnet-b0	val	0.2884	0.6572	0.7587
U-Net++ Attn	timm-effnet-b0	train	0.3020	0.6422	0.7465
U-Net++ Attn	timm-effnet-b0	val	0.2910	0.6558	0.7566
MANet	timm-effnet-b0	train	0.3162	0.6446	0.7480
MANet	timm-effnet-b0	val	0.3020	0.6555	0.7602
SegFormer	timm-effnet-b0	train	0.2719	0.6686	0.7713
SegFormer	timm-effnet-b0	val	0.2889	0.6540	0.7582
TransUNet	h=8, d=6, mlp=5	train	0.6176	0.5121	0.6217
TransUNet	h=8, d=6, mlp=5	val	0.5710	0.5736	0.6879
Swin U-Net	mit-b2	train	0.4385	0.5273	0.6325
Swin U-Net	mit-b2	val	0.3740	0.5824	0.6888

Overall, convolutional architectures consistently outperform transformer-based models in this task. Among CNNs, U-Net++, MANet, and among transformer-based models SegFormer achieve the highest IoU values, indicating strong capability in delineating lake boundaries across diverse radiometric conditions. Other transformer-based models, including TransUNet and Swin U-Net, exhibit lower segmentation accuracy, particularly for small and fragmented lakes, suggesting that their higher model complexity may not be well matched to the available training data volume.

Extended training of the top-performing RGB models further improved generalization performance. As shown in Table II, MANet achieved the highest validation IoU, followed closely by U-Net++ and SegFormer. Based on this result, MANet was selected for subsequent inventory generation and downstream analysis.

B. Impact of Spectral Configuration

To assess whether additional spectral information improves segmentation performance, we trained the top-performing models using multi-band inputs, including the full 8-band surface reflectance product and a hybrid RGB+NIR+NDWI configuration. Results are summarized in Table III.

TABLE II
FINAL 20-EPOCH RESULTS ON RGB

Model	Split	Loss	IoU	F1
UNet++	train	0.2813	0.6609	0.7606
UNet++	val	0.2825	0.6652	0.7663
MANet	train	0.2318	0.7127	0.8043
MANet	val	0.2685	0.6770	0.7770
SegFormer	train	0.2143	0.7232	0.8186
SegFormer	val	0.2826	0.6607	0.7634

TABLE III
PERFORMANCE WITH MULTI-BAND INPUTS

Model	Bands	Split	Loss	IoU	F1
UNet++	8-band	train	0.3279	0.6178	0.7284
UNet++	8-band	val	0.3140	0.6361	0.7364
MANet	8-band	train	0.3172	0.6324	0.7415
MANet	8-band	val	0.3227	0.6371	0.7347
SegFormer	8-band	train	0.3250	0.6158	0.7298
SegFormer	8-band	val	0.3059	0.6440	0.7443
UNet++	RGB+NIR+NDWI	train	0.2958	0.6437	0.7513
UNet++	RGB+NIR+NDWI	val	0.3491	0.5946	0.7013
MANet	RGB+NIR+NDWI	train	0.2979	0.6423	0.7487
MANet	RGB+NIR+NDWI	val	0.3855	0.5625	0.6690
SegFormer	RGB+NIR+NDWI	train	0.2867	0.6508	0.7605
SegFormer	RGB+NIR+NDWI	val	0.3379	0.6042	0.7052

Contrary to expectations, incorporating additional spectral bands did not lead to consistent performance gains. This suggests that the spectral information most relevant for glacial lake discrimination is already captured in visible bands, while additional channels may introduce noise or increase optimization difficulty given the limited training data. As a result, RGB-only inputs were used for final inventory generation.

C. Glacial Lake Inventory Changes Between 2020 and 2025

Using the automated deep learning framework, a regional glacial lake inventory was generated for the year 2025 and compared with the 2020 baseline inventory. At the regional scale, both the number of lakes and the total lake area increased over the five-year period. The 2020 inventory contains 8,808 glacial lakes with a total surface area of 136.399 km², whereas the 2025 inventory contains 15,353 lakes covering 148.613 km².

This corresponds to a net increase of 6,545 lakes and a total area gain of 12.214 km². The percentage area change (PAC) over the 2020–2025 period is 8.95%, equivalent to an annual percentage area change (APAC) of 1.79% yr⁻¹. These results indicate sustained glacial lake expansion across the region, driven primarily by the formation of new lakes, while also highlighting ongoing areal growth at the regional scale.

D. Size-Class Dependent Changes

To examine how glacial lake evolution varies with lake size, changes in lake number and total area were analyzed across predefined size classes (Table IV). The increase in lake count is overwhelmingly dominated by the smallest lakes (< 0.01 km²), which account for 6,454 of the 6,545 newly identified lakes between 2020 and 2025. This size class also contributes a net area gain of 5.73 km².

Lakes in the 0.1–0.5 km² size range exhibit the largest absolute area increase (8.06 km²), indicating active expansion

among medium-sized lakes. In contrast, the 1–5 km² size class shows a net area loss of 5.62 km² despite only a small reduction in lake count, suggesting shrinkage or partial drainage of larger lakes. These results highlight a heterogeneous pattern of lake evolution, characterized by widespread formation of small lakes alongside contrasting trends among larger systems.

TABLE IV
CHANGES IN GLACIAL LAKE NUMBER AND AREA BY SIZE CLASS
BETWEEN 2020 AND 2025.

Size (km ²)	N_{2020}	A_{2020} (km ²)	N_{2025}	A_{2025} (km ²)	ΔN	ΔA (km ²)
< 0.01	6,752	13.99	13,206	19.72	+6,454	+5.73
0.01–0.05	1,513	35.42	1,518	35.15	+5	−0.27
0.05–0.1	302	20.87	347	24.23	+45	+3.36
0.1–0.5	218	39.17	259	47.23	+41	+8.06
0.5–1	14	10.02	16	10.97	+2	+0.95
1–5	9	16.94	7	11.32	−2	−5.62

E. Elevation-Band Distribution of Changes

Changes in glacial lake distribution were further analyzed across 500 m elevation bands (Table V). The majority of lakes in both inventories are concentrated between 4,000 and 5,000 m elevation. However, this elevation range also exhibits the largest net area loss, particularly within the 4,000–4,500 m band, which shows a reduction of 9.00 km².

In contrast, elevation bands below 4,000 m and above 4,500 m generally experience net increases in both lake count and total area. Notably, the 5,500–6,000 m band shows a substantial area gain despite containing relatively few lakes, reflecting the emergence or rapid growth of high-altitude lakes. These patterns suggest elevation-dependent controls on glacial lake formation and stability across the HKH region.

TABLE V
CHANGES IN GLACIAL LAKE NUMBER AND AREA BY ELEVATION BAND
BETWEEN 2020 AND 2025.

Elevation (m)	N_{2020}	A_{2020} (km ²)	N_{2025}	A_{2025} (km ²)	ΔN	ΔA (km ²)
3000–3500	459	5.86	1,781	7.55	+1,322	+1.69
3500–4000	2,142	29.42	2,818	30.47	+676	+1.05
4000–4500	4,089	63.56	3,709	54.56	−380	−9.00
4500–5000	1,835	28.07	2,709	30.93	+874	+2.86
5000–5500	207	1.55	576	2.00	+369	+0.45
5500–6000	4	0.02	94	2.62	+90	+2.60
> 6000	0	0.00	12	0.01	+12	+0.01

F. Lake-Level Change Classification

Lake-level dynamics were analyzed using an overlap-based polygon matching approach between the 2020 and 2025 inventories. Results show that newly formed lakes dominate the 2025 inventory, with 12,778 lakes classified as new. Among matched lakes, 1,331 remain stable, while 836 exhibit expansion and 408 show shrinkage. This distribution confirms that the observed increase in lake count is primarily driven by lake formation rather than systematic expansion of pre-existing lakes, underscoring the highly dynamic nature of glacial lake evolution in the region.

G. Hazard Classification and Identification of PDGLs

Applying the multi-factor hazard assessment framework to the 2025 inventory yields a broad distribution of hazard levels across the mapped glacial lakes. Lakes classified as medium hazard form the largest group comprising 4,638 lakes, followed by low and high hazard classes which have 4,101 and 3,040 lakes, respectively. A total of 1,479 lakes fall into the very high hazard category.

Lakes classified as high or very high hazard are designated as potentially dangerous glacial lakes (PDGLs). Based on this criterion, 4,519 lakes are identified as PDGLs, representing a substantial fraction of the total lake population. The large number of PDGLs highlights the importance of region-wide hazard screening approaches to prioritize sites for detailed investigation and risk mitigation.

H. Downstream Population Exposure

Downstream population exposure associated with potentially dangerous glacial lakes was estimated by aggregating gridded population counts within cumulative distance bands. Across all PDGLs, approximately 1.58×10^7 people reside within 5 km downstream, increasing to 7.78×10^7 within 10 km and 4.15×10^8 within 20 km. These results indicate that a large population may be exposed to GLOF-related hazards and underscores the societal relevance of automated glacial lake mapping and hazard assessment frameworks for disaster preparedness and regional risk management.

IV. CONCLUSION AND OUTLOOK

This study presents an automated deep learning-based framework for regional-scale glacial lake mapping and hazard assessment across the Hindu Kush–Himalaya using high-resolution PlanetScope imagery. Systematic evaluation of convolutional and transformer-based models shows that RGB-based convolutional architectures provide the most robust performance for operational inventory generation. Application of the framework yields an updated 2025 glacial lake inventory and reveals continued lake development, characterized by increasing lake numbers, net area expansion, and heterogeneous size- and elevation-dependent changes. By integrating a multi-factor hazard assessment, a substantial fraction of lakes are identified as potentially dangerous, highlighting the importance of moving beyond inventory-based analyses toward risk-informed assessments. Downstream population exposure estimates further underscore the potential societal impacts of glacial lake hazards. The proposed framework offers a scalable and repeatable approach for regional hazard screening in data-scarce high mountain regions. Future work will focus on improving temporal resolution, incorporating additional cryospheric and climatic variables, and linking automated lake detection with physically based flood modeling to support more actionable risk assessments.

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